

S002 - Batch Integration Requirements Specification

Requirements Document | Engineering

Document ID	PRJ-S002-001	Seller ID	S002
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2024-10-08
Related Scenario ID	Not scenario-specific	Document Type	Requirements Document

1. Purpose and Audience

This specification defines the requirements for BluePeak Electronics (S002) batch integration with Nexora Marketplace, as revised for the two-centre fulfilment model introduced in November 2024. It is the reference document for how catalogue, price, stock and order data move between BluePeak and the platform, and for the service expectations attached to each flow.

The audience is platform engineering, operations and BluePeak's integration team.

2. Why Batch

BluePeak lists approximately 142,000 SKUs. Prices change daily across a large part of the catalogue and stock moves continuously across two fulfilment centres. A per-movement real-time integration of the kind used by S001 was evaluated and rejected for this account on cost grounds: the call volume would have been roughly two orders of magnitude higher than the platform's next largest seller, with no corresponding commercial benefit.

The accepted trade-off is a freshness window. Batch integration means published data is correct as of the last successful run, not as of now. Every requirement in this document exists to make that window predictable and bounded.

3. Functional Requirements

3.1 Catalogue Feed

ID	Requirement	Priority
CAT-1	BluePeak shall deliver a full catalogue file daily by 23:00 IST	Must
CAT-2	The file shall be UTF-8 encoded CSV with a fixed column contract	Must
CAT-3	Each row shall carry a stable seller SKU identifier	Must
CAT-4	Rows failing validation shall be rejected individually, not by file	Must
CAT-5	A rejection report shall be available within 30 minutes of processing	Should
CAT-6	Delta files may be delivered between full files	Could

3.2 Price Feed

ID	Requirement	Priority
PRC-1	Price changes shall be delivered in a dedicated file, separate from catalogue	Must
PRC-2	Price files shall be processed within 45 minutes of delivery	Must
PRC-3	A price change greater than 40 percent shall be held for confirmation	Must
PRC-4	Price and stock files shall be independently processable	Must

Requirement PRC-3 exists because a supplier feed error in 2023 published a set of accessories at a fraction of cost. The hold threshold has been triggered nineteen times since; in seventeen cases it was a genuine error.

3.3 Stock Feed

ID	Requirement	Priority
STK-1	Stock shall be delivered per fulfilment centre, not aggregated	Must
STK-2	The stock window shall be computed in Asia/Kolkata	Must
STK-3	Published availability shall be no more than 6 hours behind the source	Must
STK-4	SKU mapping shall be resolved against the current catalogue version	Must
STK-5	A failed run shall alert operations within 15 minutes	Must
STK-6	Zero-stock rows shall be explicit, not omitted	Must

Requirements STK-2, STK-3 and STK-4 were tightened in June 2026 following the May 2026 synchronisation drift, where a fixed UTC offset in the scheduler and a stale mapping table together produced a 26-hour drift. The original specification assumed correct scheduling rather than requiring it to be verified.

3.4 Order Flow

ID	Requirement	Priority
ORD-1	Orders shall be retrievable through the Order API	Must
ORD-2	Order state changes shall be consumable from the event bus	Must
ORD-3	Consumers shall tolerate at-least-once delivery	Must
ORD-4	Consumer lag shall be monitored and alerted above 15 minutes	Must

ID	Requirement	Priority
ORD-5	Order retrieval shall support pagination with stable ordering	Must

Requirement ORD-4 was added in July 2026 after a consumer rebalance loop left 6,700 events unprocessed and the downstream settlement feed nine hours stale.

4. Non-Functional Requirements

ID	Requirement	Target
NFR-1	Catalogue file processing throughput	142,000 rows within 90 minutes
NFR-2	Stock file processing throughput	Full stock position within 25 minutes
NFR-3	Order API availability	99.9 percent monthly
NFR-4	Order API p99 latency under normal load	Under 900 ms
NFR-5	Order API behaviour under overload	Shed load with 429 and Retry-After, not 503
NFR-6	Credential rotation	Automated, maximum age 90 days
NFR-7	Transport security	SFTP over SSH with key authentication and IP allowlist

NFR-5 is a direct consequence of the July 2026 flash sale incident, where an unbounded client retry policy and a missing circuit breaker turned a capacity problem into a cascading failure. Graceful load shedding is now an explicit requirement rather than an assumption.

5. Data Quality Rules

1. A SKU present in the stock file but absent from the current catalogue version shall be rejected and reported, never silently mapped.
2. A catalogue row whose brand or category is unrecognised shall be rejected.
3. A stock quantity that is negative shall be treated as zero and reported.
4. A price of zero shall be rejected.
5. A file whose row count differs from the previous run by more than 25 percent shall be held for operator confirmation.

Rule 1 is the control that addresses the stale mapping table contributing factor from May 2026. Rule 5 is a long-standing control that has prevented several truncated-file incidents.

6. Operational Requirements

Area	Requirement
Scheduling	All jobs pinned to an explicit timezone; no reliance on host locale
Monitoring	Freshness, lag and rejection rate exposed as operational metrics
Alerting	Operations alerted on run failure, freshness breach and consumer lag
Capacity	10 working days notice required before a promotional flash sale
Access	SFTP service accounts only; no interactive credentials

Area	Requirement
Runbooks	Service Recovery Procedure applies to all feed failures

7. Acceptance

Acceptance of any change to these flows requires a 72-hour parallel run with zero unexplained variance in published stock, a load test at three times observed peak for API changes, and written confirmation from both the platform engineering owner and BluePeak's integration lead.

8. Revision History

Version	Date	Change
1.0	2022-05-12	Initial specification for single-centre batch integration
2.0	2024-10-08	Revised for two fulfilment centres; per-centre stock required
2.1	2026-06-08	STK-2, STK-3, STK-4 tightened after synchronisation drift
2.2	2026-07-28	NFR-5 and ORD-4 added after order API and consumer incidents